IBRAHIM HOSSAIN SAJAL

ihossain2@isrt.ac.bd | (682) 232-8791 | 9609 Medical Center Drive, Rockville, MD 20850

EDUCATION

2024 **Ph.D.** in Data Science & Statistics, University of Texas at Dallas, Richardson, TX
2018 **M.S.** in Applied Statistics, University of Dhaka, Dhaka, Bangladesh
2017 **B.S.** in Applied Statistics, University of Dhaka, Dhaka, Bangladesh

PROFESSIONAL EXPERIENCE

2025 **Postdoctoral Fellow**, Integrative Tumor Epidemiology Branch, Division of Cancer Epidemiology and Genetics, National Cancer Institute, Rockville, MD
2019-2024 **Graduate Teaching Assistant**, Department of Mathematical Sciences, University of Texas at Dallas, Richardson, TX
2019 **Sr. Statistician**, James P Grant School of Public Health, BRAC University, Dhaka, Bangladesh
2018-2019 **Statistician**, James P Grant School of Public Health, BRAC University, Dhaka, Bangladesh
2017-2018 **Statistical Intern**, James P Grant School of Public Health, BRAC University, Dhaka, Bangladesh

TEACHING EXPERIENCE

2026 **Instructor**, "Epidemiology 101: Causality and Intro to Statistics", Division of Cancer Epidemiology and Genetics, National Cancer Institute, Rockville, MD

2019-2024 **Graduate Teaching Assistant**, Department of Mathematical Sciences, University of Texas at Dallas, Richardson, TX
Co-Instructor
MATH 2413: Differential Calculus — Spring 2021, Fall 2020, Fall 2024
MATH 2414: Integral Calculus — Fall 2022, Spring 2022, Fall 2021
MATH 2417: Calculus I — Fall 2023, Fall 2019
MATH 2418: Linear Algebra — Spring 2023
Grader
MATH 1314: College Algebra — Fall 2023
STAT 2332: Introductory Statistics for Life Sciences — Spring 2021, Fall 2020, Summer 2020, Fall 2024
CS 3341: Probability and Statistics in Computer Science — Fall 2022, Summer 2021, Spring 2020
STAT 4354: Numerical Statistical Computing — Spring 2024
STAT 5352: Probability and Statistics II — Spring 2020
STAT 6341: Numerical Linear Algebra and Statistical Computing — Fall 2021
STAT 6348: Applied Multivariate Statistics — Spring 2024

2018 **Instructor and Course Coordinator**, "Orientation to STATA," James P Grant School of Public Health, BRAC University, Dhaka, Bangladesh

COMPETITIVE RESEARCH FUNDING

2023 Travel Award, Advances in Statistical and Computational Methods for Analysis of Biomedical, Genetic, and Omics Data, Dallas, TX

2022 Boyd Harshbarger Travel Award, Southern Regional Conference on Statistics, Jekyll Island, GA

2022 PhD Research Small Grant Program, Office of Graduate Education, University of Texas at Dallas, Richardson, TX

AWARDS, FELLOWSHIPS, AND SCHOLARSHIPS

2026 DCEG Fellows Award for Research Excellence (DFARE), Division of Cancer Epidemiology and Genetics, National Cancer Institute, Rockville, MD

2025 Fellows Award for Research Excellence (FARE), National Institutes of Health, Bethesda, MD

2023, 2022 Mei Lein Fellowship, School of Natural Science and Mathematics, University of Texas at Dallas, Richardson, TX

2021 Summer Institute in Statistical Genetics Scholarship, University of Washington, Seattle, WA

2019-2024 Graduate Tuition Scholarship, Department of Mathematical Sciences, University of Texas at Dallas, Richardson, TX

2018 Academic Excellence Award, University of Dhaka, Dhaka, Bangladesh

INVITED PRESENTATIONS

2025 “Multivariate Quantitative Bayesian LASSO for detecting rare haplotype association with correlated continuous phenotypes,” Annual Joint Statistical Meeting (JSM) on Statistics, Data Science, and AI Enriching Society, Nashville, TN

POSTER PRESENTATIONS

2026 “Integrative analysis identifies proteomic intermediates associated with renal cell carcinoma and its risk factors,” American Association for Cancer Research, San Diego, CA

2026 “Meta-trans: A generalized summary statistics-based aggregative approach to identify trans-regulation of gene-sets by disease related variants,” STATGEN: Conference on Statistics in Genomics and Genetics, Atlanta, GA

2025 “Integrative analysis identifies proteomic intermediates associated with renal cell carcinoma and its risk factors,” NIH Research Fair, Bethesda, MD

2023 “Bivariate Quantitative Bayesian LASSO for detecting rare haplotype association with correlated continuous phenotypes,” Advances in Statistical and Computational Methods for Analysis of Biomedical, Genetic, and Omics Data, Dallas, TX

2022 “Bivariate Quantitative Bayesian LASSO for detecting rare haplotype association with correlated continuous phenotypes,” Southern Regional Council on Statistics, Jekyll Island, FL

BIBLIOGRAPHY

1. **Sajal IH**, Song AJ, Brown KM, Machiela MJ, Kraft P, Chanock S, Purdue MP, Dutta D. Integrative analysis prioritizes proteins associated with renal cell carcinoma and its risk factors. JNCI Cancer Spectrum. 2026 Jul 13:pkag074. doi: 10.1093/jncics/pkag074. Epub ahead of print. PMID: 42439544.
2. **Sajal IH**, Pfeiffer RM, Jatoi I, Gail MH, Cecchini RS, Choudhary PK, Biswas S. CBCRisk-Mastectomy: A Prediction Tool to Aid Contralateral Prophylactic Mastectomy Decision- making. Journal of Surgical Research. 2026 Jun 23;325:336-346. doi: 10.1016/j.jss.2026.04.040. Epub ahead of print. PMID: 42335630.
3. **Sajal IH**, Biswas S. Bivariate quantitative Bayesian LASSO for detecting association of rare haplotypes with two correlated continuous phenotypes. Frontier in Genetics. 2023 Mar: 9;14:1104727. PMID: 36968609; PMCID: PMC10033866.
4. **Sajal IH**, Chowdhury M, Wang T, Euhus D, Choudhary PK, Biswas S. CBCRisk-Black: a personalized contralateral breast cancer risk prediction model for black women. Breast Cancer Research & Treatment. 2022:194(1):179-186. Epub 2022 May 13. PMID: 35562619; PMCID: PMC9637387.
5. Talukder A, Gupta RD, Hashan MR, Haider SS, **Sajal IH**, Sarker M. Association between television viewing and overweight and obesity among women of reproductive age in Timor-Leste: evidence from the demographic health survey 2016. BMJ Open 2021 Aug: 11;11(8):e045547. PMID: 34380718; PMCID: PMC8359523.
6. Gupta RD, Akonde M, **Sajal IH**, Al Kibria GM. Association between height and hypertension among US adults: analyses of National Health and Nutrition Examination Survey 2007-18. Clinical Hypertension. 2021 Feb: 26;27(1):6. PMID: 33637121; PMCID: PMC7908753.
7. Gupta RD, Haider SS, Hashan MR, Hasan M, Sutradhar I, **Sajal IH**, Joshi H, Haider MR, Sarker M. Association between the frequency of television watching and overweight and obesity among women of reproductive age in Nepal: Analysis of data from the Nepal Demographic and Health Survey 2016. PLoS One 2020 Feb: 10;15(2):e0228862. PMID: 32040537; PMCID: PMC7010261.
8. Gupta RD, Haider SS, Sutradhar I, Hashan MR, **Sajal IH**, Hasan M, Haider MR, Sarker M. Association of frequency of television watching with overweight and obesity among women of reproductive age in India: Evidence from a nationally representative study. PLoS One 2019 Aug: 29;14(8):e0221758. PMID: 31465465; PMCID: PMC6715273.
9. Gupta RD*, **Sajal IH***, Hasan M, Sutradhar I, Haider MR, Sarker M. Frequency of television viewing and association with overweight and obesity among women of the reproductive age group in Myanmar: results from a nationwide cross-sectional survey. BMJ Open 2019 Mar: 20;9(3):e024680. PMID: 30898812; PMCID: PMC6475159. *Authors contributed equally as first authors

MANUSCRIPTS UNDER SUBMISSION

1. **Sajal IH**, Shi J, Kraft P, Dutta D. HARMONI: heterogeneity-aware inference for molecular association studies of diseases.
2. **Sajal IH**, Wang K, Zhang H, Dutta D. Multi-Trans: A generalized summary statistics-based aggregative approach to identify trans-regulation of gene-sets by disease related variants.

MANUSCRIPTS IN PREPARATION

1. **Sajal IH**, Biswas S. Multivariate Quantitative Bayesian LASSO with MFVB-based fast posterior approximation for detecting rare haplotype associations with correlated continuous phenotypes.
2. **Sajal IH**, Purdue M., Dutta, D. Genetic and Observational Evidence Linking BMI to Survival in Renal Cell Carcinoma.

SPECIALTY TRAINING

2026 Human Microbiome and Cancer Epidemiology, Division of Cancer Epidemiology and Genetics, National Cancer Institute, Rockville, MD

2026 DCEG Presentation Workshop, Division of Cancer Epidemiology and Genetics, National Cancer Institute, Rockville, MD

2022 Python for Data Science, University of Texas at Dallas, Richardson, TX

2021 Introduction to Genetics and Genomics, Summer Institute in Statistical Genetics, University of Washington, Seattle, WA

2021 Genetic Epidemiology, Summer Institute in Statistical Genetics, University of Washington, Seattle, WA

2021 MCMC for Genetics, Summer Institute in Statistical Genetics, University of Washington, Seattle, WA

SERVICE TO THE PROFESSION

Service

2026 Judge, Postbaccalaureate Poster Day, National Institutes of Health, Bethesda, MD

2026 Judge, FARE, National Institutes of Health, Bethesda, MD

Peer Reviewer

2026 Journal of International Medical Research

2026 Scientific Reports

2025 Communications in Statistics - Simulation and Computation

PROFESSIONAL SOCIETIES

2026-present American Association for Cancer Research

2021-present American Statistical Association

MEDIA COVERAGE

2023 "UTD Researchers Develop Tool to Help Black Breast Cancer Survivors." School of Natural Sciences and Mathematics, The University of Texas at Dallas, June 12, 2023.

2022 "A New Tool from UT Dallas Addresses Racial Disparities in Breast Cancer Patients." KERA News, October 28, 2022.

2022 "UTD Researchers Address Racial Disparity in Breast Cancer Risk Assessment." University of Texas at Dallas Magazine, October 20, 2022.

VOLUNTEER EXPERIENCES

2026 Shepherd's Table at Silver Spring, MD. Serving dinner to people experiencing homelessness and food insecurity.

2026, 2025 The Children's Inn at NIH. Preparing and serving dinner to cancer patients and families.

2023 Advances in Statistical and Computational Methods for Analysis of Biomedical, Genetic, and Omics Data. Dallas, TX

2022, 2021 Florence Nightingale Day (Event promoting statistics to high school students), Dallas, TX

2019 Global Conference on Implementation Science (GCIS) Dhaka, Bangladesh

2014 International Conference on Applied Statistics (ICAS) Dhaka, Bangladesh